Operating System

(41TRC2)

IT IV Semester

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# Lab Assignment 3

Aim: To create shell scripts for the following questions.
To Perform: To code and solve the following problems.
To Submit: Provide shell scripts for the following:

1. Find the Largest of Three Numbers.

   echo "Enter three numbers: " read a b c if [ $a -ge $b ] && [ $a -ge $c then echo "Largest number is $a" elif [ $b -ge $a ] && [ $b -ge $c ]; then echo "Largest number is $b" else echo "Largest number is $c" fi

2. Check if a Year is a Leap Year. #!/bin/bash echo "Enter a year:" read year if (year % 4 O year % 100 =O) I I (year % 400 0) )); then echo "$year is a leap year." else echo "$year is not a leap year."

3. Check if Triangle is Valid.

   #!/bin/bash

   echo "Enter three angles of a triangle:" read a b c sum=$((a + b + c)) if [ $sum -eq 180 ]; then echo "It is a valid triangle."
   else echo "It is not a valid triangle." fi

4. Check Character Type.

   #!/bin/bash

   echo "Enter a character:" read char if "$char" [a-zA-Z] I];
   then echo "It is an alphabet."
   elif "$char ' ' - -N [0-9] then echo
   "It is a digit." else echo "It is a special character."

5. Calculate Profit or Loss. #!/bin/bash echo "Enter Cost Price:" read cp echo "Enter Selling Price:" read sp if [ $sp -gt $cp ]; then echo "Profit: $(($sp $cp))"elif [ $cp -gt $sp l; then echo "Loss: $(($cp - $sp))" else echo "No Profit, No Loss." fi

6. Print Even and Odd Numbers from 1 to 10.

```
#!/bin/bash echo "Even
numbers:" for i in {2..10..2}; do
echo $i; done echo "Odd
numbers:" for i in {1..9..2}; do
echo $i; done
```

7. Print Multiplication Table #!/bin/bash echo "Entera number:" read n

```
for i in {1..10}; do echo
"$n x$i = $((n * i))" done
```

8. Factorial of a Number. #!/bin/bash echo "Entera number:" read n fact-I for (( i=l; i<=n; i++ )); do fact=$((fact * i)) done echo "Factorial of $n is $fact"

9. Sum of Even Numbers from 1 to 10.

```
#!/bin/bash

sum=O for i in {2..10..2}; do sum=$((sum + i))
done echo "Sum of even numbers from 1 to 10 is
$sum"
```

10. Sum of Digits of a Number.

```
#!/bin/bash echo
"Entera number:" read
num
sum=O while [ $num -gt 0 ];
do sum=$((sum + num %
10)) num=$((num / 10))
done echo "Sum of digits is
$sum"
```

11. Basic Calculator. #!/bin/bash echo "Enter two numbers:" read a b

```
echo "Choose operation: +- * /"
read op case $op in
   "+ 11 ) echo "Result: $((a + b))" ;;
   „- " ) echo "Result: $((a - b))" ;;
   "*") echo "Result: $((a * b))" ;;
   "/") echo "Result: $((a / b))" ;;
   *  )  echo  "Invalid  operation"  „
Esac
```

## 12. Print Days of the Week

```
#!/bin/bash    days=("Sunday"    "Monday"    "Tuesday"    "Wednesday"
"Thursday" "Friday"
"Saturday")  for  day  in  "${days[@]}";  do
echo $day; done
```

## 13. Print First 4 Months with 31 Days

```
#!/bin/bash
months=("January" "March" "
for month in "${months[@]}";
for month in
```

```
"March"          "May"
"July")      do      echo
$month; done
```

## 14. Using Functions

## (a) Check Armstrong Number

```
#!/bin/bash
is_armstrong()    {    num=$l
  sum=0  n=${#num}  for  ((
  i=0;      i<n;      i++      do
  digit=${num:i:l}
  sum=$((sum  +  digit*  *n))
  done
  [[$sum -eq $num ]] && echo "$num is an Armstrong number" I I echo
"$num is not an Armstrong number"

echo  "Entera  number:"
read  num  is  armstrong
```

```
$num (b) Check
Palindrome #!/bin/bash
is_palindrome() {
num=$l rev=$(echo
$num I rev)
  [[$num -eq $rev ]] && echo "$num is a palindrome" I l echo ''$num is
not a palindrome"

echo "Entera number:"
read num
is_palindrome $num
(c) Fibonacci Series
#!/bin/bash
fibonacci() { a=0b=l echo -
  n "$a $b" for (( i=2; i<$l;
  i++ )); do c=$((a + b))
  echo -n " $c" a=$b b=$c
  done echo

echo "Enter number of terms:" read n fibonacci
$n (d) Check Prime or Composite #!/bin/bash
is_prime() { num=$l if [ $num -It 2 ]; then echo
"$num is neither prime nor composite" return fi
for (( i=2; i*i<=num; i++ )); do if [ $((num % i)) -eq
O then echo 'I $num is composite" return
     fi done echo
  "$num is prime"

echo "Entera number:" read num
is_prime $num (e) Convert
Decimal to Binary #!/bin/bash
decimal_to_binary() { num=$l
binary= """while [ $num -gt O l; do
binary=$((num % 2))$binary
num=$((num / 2)) done echo
"Binary: $binary"
```

```
echo "Enter a decimal number:"
read   num   decimal_to_binary
$num
```